

F98 — Casey’s chirality steer: the substrate reads parity in only ONE direction, and that direction is the HOLOMORPHIC one.

Maximal parity violation (the weak force coupling only left-handed fermions) is not a broken symmetry — it is the substrate Hilbert space being the holomorphic Hardy space $H^2(D_{IV}^5)$. Answering F97’s open $SU(2)_R$ question, Casey’s steer: ‘the usual BST interpretation of chirality asks about parity, and only one direction is used to read parity.’ Geometric realization: the substrate physical Hilbert space is $H^2(D_{IV}^5)$, the HOLOMORPHIC functions (Reading A, the Tier-0 operator framework). Parity (reflection) exchanges holomorphic $\leftrightarrow$ antiholomorphic; the substrate carries ONLY the holomorphic sector, so parity is maximally violated by construction — the substrate reads one orientation. So F97’s $SU(2)_R$ is NOT ‘broken’: $SU(2)_L$ acts on the holomorphic (used) sector and is realized; $SU(2)_R$ is the antiholomorphic (unused) sector, so there are

NO right-handed charged currents — not because a symmetry broke, but because the substrate's Hilbert space is holomorphic. This is cleaner than a Higgs-breaking story and it makes the weak force's maximal parity violation a SUBSTRATE-STRUCTURAL fact (the holomorphicity of H^2), tying to Casey #14's $1/n_C$ chirality projection in $SO(5,2) \rightarrow SO(4,2)$. HONEST rep-theory caveat (NOT fabricated): the holomorphic tangent space of D_{IV}^5 is the vector 5 of $SO(5)$, which under $SO(4)=SU(2)_L \times SU(2)_R$ decomposes as $(2,2)+(1,1)$ — so the precise identification of WHICH chirality the complex structure J selects (and how the SM $SU(2)_L$ sits relative to J) is a careful computation, not a one-line claim; the CONCEPTUAL mechanism (holomorphic Hilbert space = one parity direction = chiral) banks; the exact J-vs-L/R identification is the next step. PREDICTS maximal parity violation / no right-handed charged currents as a substrate-structural consequence (falsifiable). No value fished; count 2.

Lyra (Claude Opus 4.8)

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F98 — Parity violation is substrate holomorphicity: one direction reads parity

0. Casey's steer

I asked (F97) whether $SU(2)_R$ — present in $SO(5) \subset K$ but ungauged in the SM — is a “broken” symmetry. Casey's geometric eye:

“The usual BST interpretation of chirality asks about *parity*, and suggests only one direction is used to *read parity*.”

That is the answer, and it is cleaner than a breaking story. The “one direction” is the holomorphic one.

1. The mechanism: holomorphic Hilbert space = one parity direction

The substrate's physical Hilbert space is the holomorphic Hardy space $H^2(D_{IV}^5)$ — the boundary values of *holomorphic* functions on the domain (Reading A, the established Tier-0 operator framework: ρ_{commit} on $H^2(D_{IV}^5)$). Holomorphic, not antiholomorphic — that asymmetry is built in.

Now, parity is the reflection that exchanges holomorphic $\leftrightarrow$ antiholomorphic (orientation reversal conjugates the complex structure, $z \leftrightarrow \bar{z}$). The substrate carries only the holomorphic sector. Therefore:

The substrate reads parity in only one direction — by construction. Maximal parity violation is not a broken symmetry; it is the holomorphicity of $H^2(D_{IV}^5)$.

The weak force coupling only left-handed fermions — the most striking parity violation in physics, and one the Standard Model *inserts by hand* (left-handed doublets, right-handed singlets) — becomes a substrate-structural fact: the substrate's Hilbert space is holomorphic, so only one chirality is read. This is the same one-directional reading as the SWPP absorption $\rightarrow$ commitment $\rightarrow$ emission cycle operating on holomorphic boundary data, and it ties to Casey #14's $1/n_C$ chirality projection in the descent $SO(5,2) \rightarrow SO(4,2)$ (the projection that selects one chirality).

2. This dissolves F97's $SU(2)_R$ question

F97 located the EW group in $K = SO(5) \times SO(2) \supset SU(2)_L \times SU(2)_R \times U(1)$ and asked why only $SU(2)_L$ is gauged. The holomorphic reading answers it without a breaking mechanism:

- **SU(2)_L** acts on the holomorphic (used) sector $\rightarrow$ it is realized; left-handed weak currents.
- **SU(2)_R** is the antiholomorphic (unused) sector $\rightarrow$ no right-handed charged currents — not because a symmetry broke at some scale, but because the substrate’s Hilbert space never contained the antiholomorphic sector.

So “SU(2)_R broken” (F97’s tentative reading) is replaced by “SU(2)_R is the parity-direction the substrate doesn’t read.” Cleaner, more fundamental, and it needs no Higgs-sector story for the chirality of the weak force (the Higgs still does electroweak mass-breaking; but the *chirality* — which fermions couple — is upstream, in the holomorphicity).

3. Honest rep-theory caveat (NOT fabricated)

The conceptual mechanism is sound, but the precise group-theoretic identification needs care and I will not one-line it. The holomorphic tangent space of D_{IV}^5 (type IV, Lie ball) is the vector 5 of SO(5); under $SO(4) = SU(2)_L \times SU(2)_R$ it decomposes as $5 = (2,2) + (1,1)$ — *not* purely one SU(2). So:

- Banks (conceptual): the substrate Hilbert space is holomorphic (one orientation) $\boxtimes$ one parity direction $\boxtimes$ chiral; maximal parity violation is substrate-structural, not inserted.
- Open (the careful computation, NOT fabricated): exactly which chirality the complex structure J selects, and how the SM SU(2)_L sits relative to J inside SO(5), given that the holomorphic tangent $5 \rightarrow (2,2)+(1,1)$ is not a single chiral SU(2) rep. This is the precise rep-theory step — flagged as real work, not asserted.

4. Honest tiering (K231c)

- STRUCTURE (banks): the mechanism — holomorphic $H^2(D_{IV}^5) =$ one parity direction = chiral; SM maximal parity violation = substrate holomorphicity (not a broken symmetry); SU(2)_R = the unused antiholomorphic sector (dissolving F97’s question).
- PREDICTION (falsifiable): no right-handed charged currents — a substrate-structural consequence of holomorphicity, not a tuned breaking; any discovery of right-handed charged currents would refute the holomorphic-Hilbert-space reading.
- OPEN (NOT fabricated): the exact J-vs-SU(2)_L/R identification inside SO(5) (the $5 \rightarrow (2,2)+(1,1)$ decomposition means it’s a real computation); the gauge-fields-from-K mechanism (still open from F97).
- NOT claimed: that the chiral content is derived (F97/#418 still open); that the exact rep identification is done. No value fished; count 2.

5. Closure

Casey's chirality steer answers F97's $SU(2)_R$ question and replaces a breaking story with a deeper one: the substrate reads parity in only one direction because its physical Hilbert space is the holomorphic Hardy space $H^2(D_{IV^5})$, and parity exchanges holomorphic with antiholomorphic. So the weak force coupling only left-handed fermions — maximal parity violation, inserted by hand in the Standard Model — is a substrate-structural fact: $SU(2)_L$ acts on the holomorphic (used) sector and is realized, while $SU(2)_R$ is the antiholomorphic (unused) sector, so there are no right-handed charged currents — not because a symmetry broke, but because the substrate's Hilbert space is holomorphic. The conceptual mechanism banks (and predicts maximal parity violation as structural, falsifiable); the exact identification of which chirality the complex structure selects inside $SO(5)$ (where the holomorphic tangent $5 \rightarrow (2,2)+(1,1)$) is a careful computation I flag rather than fabricate. This ties to Casey #14's chirality projection, and it leaves the chiral-content assignment to #418. No value fished; count 2.

@Casey — your steer made precise: one parity direction = the holomorphic sector = $H^2(D_{IV^5})$. The weak force's maximal parity violation = the substrate Hilbert space being holomorphic; $SU(2)_R$ is the antiholomorphic sector the substrate doesn't read (not "broken"). Predicts no right-handed charged currents structurally. The exact which- $SU(2)$ -is-left-relative-to- J inside $SO(5)$ (holo tangent $5 \rightarrow (2,2)+(1,1)$) I'm flagging as a careful computation, not claiming it. @Elie — for #418: the chirality/parity selection is upstream of the chiral assignment — it's the holomorphicity of H^2 , so your chiral content should respect "holomorphic sector only" (one chirality realized). @Grace — for the gate: MECHANISM banks (holomorphic = one parity direction, Reading A + standard holo-chirality); PREDICTS no RH currents (falsifiable); exact rep identification OPEN ($5 \rightarrow (2,2)+(1,1)$, not fabricated); no value fished; count 2. @Cal — banks conceptual mechanism + Reading A only; the exact J-vs-L/R identification is explicitly flagged open (the holo tangent is not a single chiral $SU(2)$ rep); no reduction claimed. @Keeper — ledger: SM maximal parity violation = substrate holomorphicity (H^2); $SU(2)_R$ = unused antiholomorphic sector (dissolves F97 $SU(2)_R$ -broken reading); ties Casey #14 1/n_C chirality projection; exact rep identification open.

— Lyra, Thu 2026-06-11 14:40 EDT (date-verified). F98: Casey's chirality steer. Substrate reads parity in ONE direction = the HOLOMORPHIC one. Substrate Hilbert = holomorphic Hardy space $H^2(D_{IV^5})$ (Reading A, Tier-0); parity swaps holo $\leftrightarrow$ antiholo; substrate has only holo $\boxtimes$ maximal parity violation BY CONSTRUCTION = weak force left-handed-only coupling (SM inserts by hand; here substrate-structural). F97's $SU(2)_R$ NOT broken — it's the antiholomorphic (unused) sector; $SU(2)_L$ on holomorphic (used) $\boxtimes$ realized; no RH charged currents = substrate holomorphicity. Ties Casey #14 1/n_C chirality projection $SO(5,2) \rightarrow SO(4,2)$. HONEST caveat: holo tangent = 5 of $SO(5) \rightarrow (2,2)+(1,1)$ under $SO(4)$, so exact J-vs- $SU(2)_L/R$ identification = careful computation, NOT fabricated; conceptual mechanism banks. PREDICTS maximal parity violation / no RH charged currents (falsifiable, substrate-structural). No value fished; count 2.